Basic Principles of Medicine 2

Module: Digestion and Metabolism Lecture 17

Histology of the Gastrointestinal Glands – Flipped Classroom

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Recommended Reading

Histology – A Text and Atlas
Pawlina 9th Edition
Chapter 18: Digestive System III Pg. 692-729
https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290

Objectives

List the extrinsic glands of the digestive system.	SOM.MK.I.BPM2.3.DM.1.HCB.1764
Give the function(s) of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1765
Give the localization of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1766
Compare exocrine vs. endocrine functions of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1767
Describe the microscopic structure of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1768
Identify the Glisson's capsule of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1769
Describe the flow of blood in the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1770
Differentiate between the classic hepatic lobule, portal lobule and liver acinus based on microscopic structure and function.	SOM.MK.I.BPM2.3.DM.1.HCB.1771
Identify the components of the portal triad.	SOM.MK.I.BPM2.3.DM.1.HCB.1772
Identify the hepatocytes of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1773
Give the function(s) of the hepatocytes of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1774
Give the localization of the hepatocytes of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1775
Describe the microscopic structure of the hepatocytes of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1776
Identify the Ito cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1777
Give the function(s) of the Ito cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1778
Give the localization of the Ito cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1779
Describe the microscopic structure of the Ito cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1780

Your Objectives will show here!

Objectives

Identify the Kupffer cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1781
Give the function(s) of the Kupffer cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1782
Give the localization of the Kupffer cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1783
Describe the microscopic structure of the Kupffer cells of the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1784
Describe the flow of bile in the liver.	SOM.MK.I.BPM2.3.DM.1.HCB.1785
Identify the perisinusoidal space (of Disse).	SOM.MK.I.BPM2.3.DM.1.HCB.1786
Give the function of the perisinusoidal space (of Disse).	SOM.MK.I.BPM2.3.DM.1.HCB.1787
Describe the microscopic structure of the perisinusoidal space (of Disse).	SOM.MK.I.BPM2.3.DM.1.HCB.1788
Describe the histological changes seen in the liver due to cirrhosis.	SOM.MK.I.BPM2.3.DM.1.HCB.1789
Describe the changes seen in fatty liver disease.	SOM.MK.I.BPM2.3.DM.1.HCB.1790
Identify the gall bladder.	SOM.MK.I.BPM2.3.DM.1.HCB.1791
Give the function of the gall bladder.	SOM.MK.I.BPM2.3.DM.1.HCB.1792
Give the localization of the gall bladder.	SOM.MK.I.BPM2.3.DM.1.HCB.1793
Describe the microscopic of the gall bladder.	SOM.MK.I.BPM2.3.DM.1.HCB.1794
Describe the effects of cholecystokinin on the gall bladder.	SOM.MK.I.BPM2.3.DM.1.HCB.1795
Give the function of the intrahepatic billiary tree.	SOM.MK.I.BPM2.3.DM.1.HCB.1796
Describe the microscopic of the intrahepatic billiary tree.	SOM.MK.I.BPM2.3.DM.1.HCB.1797

Objectives

Discuss liver necrosis in heart failure.	SOM.MK.I.BPM2.3.DM.1.HCB.1798
Give the localization of the pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1799
Give the function(s) of the pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1800
Compare exocrine vs. endocrine pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1801
Describe the histological appearance of the exocrine pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1802
Describe the microscopic structure of the pancreatic ducts.	SOM.MK.I.BPM2.3.DM.1.HCB.1803
Discuss the enzymes produced by the exocrine pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1804
Describe the effect of cholecystokinin on the exocrine pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1805
Describe the effect of secretin on the exocrine pancreas.	SOM.MK.I.BPM2.3.DM.1.HCB.1806
Describe the histological features observed in acute pancreatitis.	SOM.MK.I.BPM2.3.DM.1.HCB.1807
Describe the outcome in exocrine glands due to mutation in the gene that codes for CFTR protein.	SOM.MK.I.BPM2.3.DM.1.HCB.1808

Lecture Cases

Once you have completed the pre-read material, you can use this knowledge as well as your own research to prepare the lecture cases. These cases will prepare you to engage fully in the flipped classroom and assist you in understanding the normal and the clinical integration of GI histology. Please note that knowledge from previous modules and terms are essential to understanding the etiology of some of these cases. Additionally, you are not responsible for the exact pathophysiology of the clinical correlates. You will be building on this knowledge in subsequent lectures in this module and in subsequent terms.

Looking forward to an exciting and interactive class!

Case 1

A 50-year-old man comes to the emergency department complaining of headache and indigestion for the past 15-20 days. He is a self-claimed alcoholic, who in the past, consumed up to 1 liter of alcohol/day (in his own language). Physical examination shows **yellowing of his sclera and yellowish discoloration of his tongue**. There is **mild abdominal distention and mild splenomegaly**. There is also reddening of the palms of his hands. A CT scan shows evidence of **portal hypertension,** and a distended gallbladder filled with sludge. Laboratory studies show abnormal Liver Function Tests.

Case 1 Questions

1. What are the functions of the liver?
   a. Which functions are exocrine vs endocrine?

2. What are the functions of hepatocytes?
   a) What organelles and cytoplasmic inclusions facilitate these functions?
   b) What histological characteristic(s) are attributed to these organelles/ inclusions?

3. What surface of the hepatocyte interacts with the perisinusoidal space of Disse?
   a. What is the function of the perisinusoidal space of Disse?
   b. What are the components of the perisinusoidal space of Disse?

4. What structures form the boundaries (apexes) of the classic liver lobule?
   a. What are some distinguishing features of the contents of the portal canal?

5. What are some of the histological changes observed in liver cirrhosis?
   a. Which cells are responsible for these changes?

Case 2

A 65-year-old man comes to the emergency department complaining of a 9-day history of upper abdominal discomfort and **yellowing of the skin**. **The pain was described as a persistent dull ache and mainly located in the epigastrium**. It was not related to food consumption. His liver function tests are abnormal. CT scan of the abdomen shows space-occupying lesions in the **common hepatic duct and left hepatic duct and obstructive dilation of the intrahepatic and extrahepatic bile ducts.** Biopsy of a sample taken from the mass in the hepatic ducts shows **dysplastic changes in the epithelial cells lining these ducts.** These cells have broken through the basement membrane into the underlying connective tissue.

Case 2 Questions

1. What structures form the boundaries (apexes) of the portal lobule?

2. What is the flow of bile within the liver?

3. What surface of the hepatocyte form bile canaliculi?
   a. What are some features of bile canaliculi?

4. Given the location of the tumor, what cell type is most likely affected?
   a. What are the functions of cholangiocytes?
   b. What cellular specializations facilitate these functions?

5. What is the name, common causes, incidence and prognosis of this tumor?

Case 3

A 64-year-old woman, CR, comes to the physician complaining of a 5-month history of progressively increasing abdominal distension and bilateral leg swelling for 1 month. She has a history of chronic cough and breathlessness which worsened during the winter for the last 10 years and recurrent episodes of swelling of her feet. There is no history of diabetes mellitus, hypertension or coronary artery disease. She does not drink alcohol but is a chronic smoker of over 40 pack years. Her pulse is 70/min, respiratory rate is 26/min and her blood pressure is 100/70 mmHg. Physical examination shows **pallor, yellowing of her sclera, and bi-pedal edema. Her neck veins are engorged, and pulsatile and jugular venous pressure raised**. **Her abdomen is distended diffusely with eversion of the umbilicus and prominent veins. Splenomegaly and hepatomegaly are present**. Respiratory examination shows **bilaterally decreased chest expansion with use of her accessory muscles.** There is also **reduced breath sounds with diffuse rhonchi over both lung fields.**

Case 3 Questions

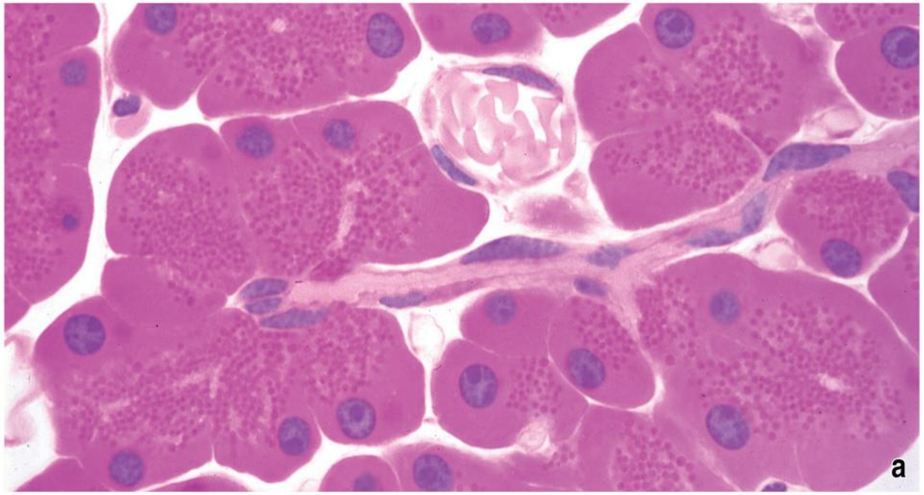
1. What is the flow of blood in the liver?
    a. What vessels supply blood to the liver?

2. What are the different zones of the functional lobule (acinus of Rappaport)?
    a. What is the implication of drug toxemia on these zones?
    b. What is the implication of hypoperfusion/hypoxemia on these zones?

3. Based on the patient's presentation, what is the most likely cause of her symptoms?
    a. What are some of the histological changes observed in liver necrosis secondary to heart failure?

Case 4

A 64-year-old woman, MG, comes to the physician because of a 1-week history of yellowing of the eyes and skin and a 3-month history of pain in the mid-abdomen that radiates to her back, itching of her skin and weight loss of 11kg (24 lbs). She has a medical history of treatment for acute inflammation of the organ shown in the image below. Physical examination shows **"scratch marks" around her body**, **temporal muscle wasting** and **yellowing of her sclera**. Abdominal imaging studies show a mass in the head of the organ in the photomicrograph below.



Case 4 Questions

1. What are the functions of the affected organ?
2. What are the histological features of the exocrine portion of the pancreas?
3. What enzymes are produced by the exocrine portion of the pancreas?
4. What hormones regulate secretion from the exocrine portion of the pancreas?
5. Given the location of the tumor, what additional digestive organs are most likely affected?
    a. What are the functions of the gall bladder?
    b. What are characteristic features of the gall bladder?
    c. What is the effect of CCK on the gall bladder?

Thank You